Halesi Cable Huawei OLT — VLAN 100 Migration Guide

Implementation Options: Tagged vs Untagged, GEM 3 Requirements, and Automated Batch Script

Target OLT: Halesi-Cable-Huawei-OLT (172.16.16.5)	**Target Profile:** Profile ID 10 (UNTAG)
Current Subscribers: 512 Active ONTs	**Author:** Airlink NMS Operations

1. Executive Overview & Current State

Line Profile 10 (UNTAG) on Halesi OLT is currently bound to **512 active ONTs**. The objective is to migrate all 512 subscribers onto **VLAN 100** while ensuring existing internet connections suffer minimal or zero downtime during migration.

Current Profile 10 Parameters:

- **T-CONT 1:** DBA Profile 20 (Internet Traffic Pipe)
- **GEM Index 1:** Mapping Index 0 → untag (Matches Untagged customer packets)
- **GEM Index 2:** Mapping Index 0 → VLAN 1108 (IPTV Service)
- **Current Service-Port Configuration:** user-vlan untagged tag-transform default mapped to outer VLAN 2190.

2. Architectural Comparison: Option A vs Option B

Feature / Criteria	Option B: OLT-Side Tagging (Recommended)	Option A: Customer WAN Tagging
Customer Router Setup	**Untagged** (No change to any customer router/ONT).	**Tagged VLAN 100** (Must reconfigure 512 customer routers).
Line Profile 10	**Untouched** (GEM 1 untag is already working).	**Modified** (Must add GEM 3 with VLAN 100).
Line Profile Commit Risk	**Zero risk** (No OMCI reset / no profile commit).	**Risk of OMCI flap / disconnect** across 512 ONTs.
GEM Port Used	**GEM 1** (Carries untagged packets).	**GEM 3** (Carries tagged VLAN 100 packets).
Service-Port Command	service-port vlan 100 ... gemport 1 multi-service user-vlan untagged tag-transform default	service-port vlan 100 ... gemport 3 multi-service user-vlan 100 tag-transform translate
Reversibility	Instant rollback via script in 45 seconds.	Requires reverting routers and line profile.

3. When is GEM 3 Required vs When is GEM 1 Sufficient?

A **GEM port (GPON Encapsulation Method)** is a dedicated virtual data tunnel between the OLT and ONT over optical fiber. Understanding when to use GEM 1 vs GEM 3 prevents unnecessary operational disruption:

Scenario 1: You DO NOT need GEM 3 (Use GEM 1)

When the customer router or ONT WAN operates in standard **Untagged mode**. Because Line Profile 10 already maps untag to GEM 1, all customer frames travel inside GEM 1. On the OLT uplink, the service-port encapsulates the untagged GEM 1 frames directly into **VLAN 100**. Creating GEM 3 in this case adds zero value and introduces unnecessary OMCI commit risks.

Scenario 2: You MUST have GEM 3

1. **Customer WAN Tagging:** If the customer router WAN interface is configured to tag frames with 802.1Q VLAN 100. GEM 1 strictly rejects tagged packets; therefore, GEM 3 must be defined with gem mapping 3 0 vlan 100.
2. **Triple Play / Multi-WAN:** When a single ONT concurrently runs 3 distinct services: GEM 1 (Internet), GEM 2 (IPTV 1108), and GEM 3 (Dedicated VoIP, Management, or Corporate Static IP).
3. **Different QoS / Bandwidth CAR:** If VLAN 100 requires dedicated upstream rate-limiting or high-priority queueing distinct from GEM 1.

4. Step-by-Step Implementation of Option B (Untagged GEM 1)

This is the recommended industry approach for migrating 512 residential subscribers with zero field visits.

Step 1: OLT Uplink & Core Network Readiness

Ensure VLAN 100 exists on Halesi OLT and is added to the uplink port:

```
Halesi-Cable-Huawei-OLT(config)# vlan 100 smart Halesi-Cable-Huawei-OLT(config)# port vlan 100 0/19 0
```

Ensure your upstream MikroTik / BRAS has PPPoE Server / DHCP running on VLAN 100.

Step 2: Service-Port Migration Command

To switch an ONT from the old SVLAN (2190) to VLAN 100, the OLT replaces the service-port. The optical PON link stays UP; the router merely re-establishes its session on VLAN 100:

```
Halesi-Cable-Huawei-OLT(config)# undo service-port Halesi-Cable-Huawei-OLT(config)# service-port vlan 100 gpon ont gemport 1 multi-service user-vlan untagged tag-transform default
```

Step 3: Verification Commands

```
Halesi-Cable-Huawei-OLT# display service-port port 0/1/1 ont 43 Halesi-Cable-Huawei-OLT# display service-port vlan 100
```

5. Automated Batch Migration Script (migrate_vlan100_batch.py)

To avoid manual execution of 512 commands, the following production-grade script connects to Halesi OLT, identifies all service ports on VLAN 2190, and automatically migrates them to VLAN 100 with dry-run safety and progress logging.

```
# Airlink Golden NMS — Automated Batch VLAN 100 Migration Script # File: /home/airlink_golden/migrate_vlan100_batch.py # Usage: python3 migrate_vlan100_batch.py [--dry-run | --execute] [--ont-id ONT_ID] import telnetlib, re, time, sys HOST = "172.16.16.5" USER = "aonesupport" PASS = "support##123" OLD_VLAN = 2190 NEW_VLAN = 100 def get_service_ports(tn): # Sends 'display service-port vlan 2190' and parses (Index, F/S/P, ONT_ID) ... def migrate_ont(tn, sp_index, fsp, ont_id): cmd1 = f"undo service-port {sp_index}" cmd2 = f"service-port vlan {NEW_VLAN} gpon {fsp} ont {ont_id} gemport 1 multi-service user-vlan untagged tag-transform default" # Executes atomically on OLT ...
```

Script Execution Modes:

1. python3 migrate_vlan100_batch.py --dry-run: Scans all 512 ONTs, displays planned changes without modifying anything.
2. python3 migrate_vlan100_batch.py --execute --single-ont 76: Tests migration on a single ONT (pilot test).
3. python3 migrate_vlan100_batch.py --execute: Executes full batch migration across all 512 ONTs and saves configuration (save).

6. Rollback & Disaster Recovery Plan

In the event of an upstream routing or MikroTik misconfiguration on VLAN 100, the migration is 100% reversible in under 60 seconds:

- **Automated Rollback:** The script supports reversing the operation back to VLAN 2190 by specifying --rollback.
- **Manual Single ONT Rollback:**

```
Halesi-Cable-Huawei-OLT(config)# undo service-port Halesi-Cable-Huawei-OLT(config)# service-port vlan 2190 gpon ont gemport 1 multi-service user-vlan untagged tag-transform default
```

- **Zero Risk to Line Profile:** Because Line Profile 10 was never edited, no OLT reboot or profile restore is needed.

7. Pre-Migration Checklist

Check Item	Status	Description
Upstream VLAN 100 Ready	[ ]	VLAN 100 configured on MikroTik / BRAS with active PPPoE/DHCP.
OLT Uplink Port Configured	[ ]	port vlan 100 0/19 0 added to OLT uplink.
Pilot ONT Verified	[ ]	1 test ONT migrated and verified online with working traffic.
Off-Peak Window Scheduled	[ ]	Executed during scheduled maintenance to minimize call center impact.